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Image Captioning Using CNN-RNN Architecture

Problem Statement:

Build a deep Recurrent Neural Network (RNN) to generate captions for images. combine a convolutional Neural Network (CNN) for feature extraction with an RNN for sequence generation.

Objectives:

* Understand image captioning using encoder decoder architecture.
* Use a pre-trained vision Transformer (ViT) as the CNN encoder and GPT-2 as the RNN decoder.
* Generate meaningful captions for visual content using beam search decoding.
* Explore the power of vision-language models for real-world application.

Scope:

This experiment demonstrates the fusion of Computer Vision and natural language processing through encoder-decoder architectures. Students gain insight into how visual features are mapped to textual sequences, component in modern AI system such as

Tools and Libraries Used:

1. Python 3.x
2. PyTorch
3. Hugging Face Transformers
4. VisionEncoderDecoderModel (ViT + GPT-2)
5. PIL (Python Imaging Library)

Implementation Steps:-

Step1: Import Required Libraries

```
import torch
from transformers import VisionEncoderDecoderModel,
                         ViTImageProcessor, AutoTokenizer
from PIL import Image
from transformers.utils import hub
hub.HUGGINGFACE_HUB_HTTP_TIMEOUT = 60
```

STEP2: Load Pretrained Image Captioning Model

```
model = VisionEncoderDecoderModel.from_pretrained
                ("nlpconnect/vit-gpt-2-image")
processor = ViTImageProcessor.from_pretrained("nlpconnect/vit-
                gpt-2-image")
device = torch.device("cuda" if torch.cuda.is_available()
                else "cpu")
model.to(device)
model.eval()
```

Step3: Define caption Generation Function

```
def generate_caption(image_path, max_length=30, num=4):
    image = Image.open(image_path).convert("RGB")

output_ids = model.generated(pixel_values, max_length=
                max_length,
                num_beams=num_beams,
                early_stopping=True)
```

Step 4: Load Image and Generate Caption

```
if __name__ == "__main__":
    img_path = "download.jpeg"
    print("Caption:", generate_caption(img_path))
```

RESULT:

The pre-trained ViT-GPT-2 model successfully generated a meaningful caption for this given image using image feature extraction and text generation.